

TigerLinkz

A Deterministic Competitive Value System for Skill-Based Digital Interaction

Abstract

Digital competitive systems increasingly rely on custodial intermediaries to enforce outcomes, allocate rewards, and manage participant balances. These intermediaries introduce discretionary control, opaque rule changes, and asymmetric risk, particularly in environments where outcomes are skill-dependent and adversarial incentives exist.

We propose TigerLinkz, a deterministic competitive system in which value transfer is directly coupled to verifiable in-game performance. Participants engage in skill-based contests governed by predefined rules, where outcomes are settled automatically according to observable game states rather than subjective arbitration. The system minimizes trust assumptions by constraining discretion, formalizing incentives, and ensuring that reward distribution is mechanically enforced.

TigerLinkz is designed under the assumption of rational, self-interested participants and operates in adversarial conditions where manipulation, collusion, and exploitation must be expected rather than prevented through policy. The system favors simplicity, verifiability, and bounded failure modes over extensibility or expressive governance.

1. Introduction

Online competitive environments have historically required trusted operators to enforce rules, resolve disputes, and distribute rewards. While effective at small scale, these systems become increasingly fragile as participant volume grows and incentives intensify. Discretionary enforcement introduces ambiguity, and ambiguity invites exploitation.

In skill-based digital competition, disputes arise not from unclear outcomes, but from unclear authority. When participants cannot independently verify results or settlement logic, trust shifts from the system itself to the operator managing it.

TigerLinkz removes this dependency by constraining outcome determination to deterministic game states and predefined settlement logic. Rather than relying on judgment, moderation, or post-hoc resolution, the system enforces outcomes through rules that execute identically for all participants.

2. Problem Definition

Competitive digital platforms suffer from three structural weaknesses:

Discretionary Settlement. Outcomes are often finalized by centralized actors who can delay, reverse, or reinterpret results.

Asymmetric Risk. Participants bear performance risk while operators retain settlement authority,

creating misaligned incentives.

Opaque Incentive Structures. Reward distribution mechanisms are frequently unclear, adjustable, or inconsistently enforced.

These weaknesses are amplified in environments involving real value transfer, where incentives to exploit ambiguity increase non-linearly with stakes.

3. System Overview

TigerLinkz is a competitive framework composed of:

A deterministic game engine that produces verifiable outcomes.

A predefined value pool committed prior to competition.

An automatic settlement mechanism that distributes value based solely on outcome data.

A constrained economic loop that discourages infinite inflation of rewards or leverage.

Participants enter contests by committing value to a match instance. Once initiated, neither participants nor operators can alter contest parameters. Upon completion, outcomes are resolved mechanically according to predefined rules. No manual intervention is required for standard operation.

4. Deterministic Outcome Enforcement

At the core of TigerLinkz is the principle that only observable, reproducible game states may influence settlement.

Game actions are evaluated in real time. Victory conditions are strictly defined. No subjective scoring or discretionary judgment exists.

If a game state can be independently reconstructed from input data, it can be used for settlement. If it cannot, it is excluded from consideration.

This constraint reduces the system's expressive flexibility but dramatically improves verifiability and trust minimization.

5. Incentive Structure

TigerLinkz assumes that participants are rational and will optimize for personal gain.

The incentive model enforces the following properties:

Symmetry. All participants are subject to identical rules.

Pre-commitment. Value is committed before outcomes are known.

Finality. Settlements occur immediately upon contest resolution.

Non-recourse. No appeals or reversals exist outside predefined failure conditions.

By removing discretionary enforcement, the system eliminates incentive vectors related to influence, negotiation, or reputation manipulation.

6. Security Considerations

TigerLinkz does not attempt to eliminate all possible attacks. Instead, it bounds failure modes.

Considered threats include collusion between participants, exploitation of game mechanics, and latency or desynchronization affecting state reproduction.

Mitigations are implemented at the rule and engine level where feasible. Where mitigation is impractical, losses are limited by predefined stake constraints rather than discretionary arbitration. The system favors predictable loss over unpredictable enforcement.

7. Limitations

TigerLinkz does not solve identity verification beyond system participation, external collusion between participants, disputes arising from non-deterministic game elements, or governance of rule evolution beyond initial constraints.

These limitations are intentional. Expanding scope would require introducing trust assumptions that conflict with the system's design goals.

8. Conclusion

TigerLinkz demonstrates that competitive digital environments can couple skill-based interaction with value transfer without relying on discretionary intermediaries. By constraining outcomes to deterministic states and enforcing settlement mechanically, the system reduces trust requirements while maintaining fairness under adversarial conditions.

Rather than attempting to eliminate risk, TigerLinkz makes risk explicit, bounded, and symmetric.

The system either functions as defined, or it does not.